

Databases and Information Systems 2026

Mandatory Assignment 1

Exercise 1. *Volt United*TM

Warning: This exercise is deliberately open. There is no single correct answer. Get creative! Also read the entire exercise before starting to solve individual subparts as some later questions might clarify some requirements for the former ones.

You work for a startup tech company known for its homonymous food delivery platform. As Volt UnitedTM's new chief data officer you are responsible for both the high-level design and the concrete implementation of the platform's data storage and processing.

The platform is accessed by three external stakeholder groups: restaurants, couriers, and customers. Customers issue food orders to restaurants, which the couriers deliver. Take-out orders (i.e., no couriers) are also supported. Food orders consist of one or more dishes. Upon order completion, customers can rate both restaurants and couriers. Customers and couriers may also tag their favorite restaurants and restaurants may tag their favorite couriers. Customers may also express cuisine preferences and restaurants cuisine specializations. All stakeholders have an associated "base" location and may decide to take part in an order based on proximity.

A salient feature of Volt UnitedTM are *united* orders from a single customer to different restaurants (delivered by a single or different couriers). This feature targets customers with incompatible food tastes within their families and friend circles. A united order demands a high degree of synchronization between the involved restaurants and couriers (to deliver warm food), which is why only selected (e.g., again proximity-based) combinations of restaurants and couriers offer united orders.

- Design a high-level E/R model for the described scenario.** Make the model detailed to faithfully represent the real world (e.g., by including relevant attributes such as dish prices or user passwords, by indicating keys, and by modeling constraints), but also do not overdo it: Your E/R diagram should easily fit on a single A4 sheet of paper (while being readable without a magnifying glass).
- Discuss your modeling choices.** How did you decide between modeling things as entities or relationships? Did you use binary or n-ary relationships (and why)? Did you use uniqueness and referential integrity constraints, weak entities, or ISA-hierarchies (and why)? Keep the arguments concise and focused on what you consider to be your model's essential aspects.
- Translate parts of your E/R model to the relational model.** Write explicit SQL `CREATE TABLE` commands for **at least three entities and two relationships** and indicate (foreign) key constraints and ways to maintain referential integrity in a meaningful way. Also provide one example instance in SQL syntax. A handful of example entities/relationships per entity/relationship set suffice.

Exercise 2. *Regular Expressions and Context Free Grammars*

- Give a natural description of the languages of the following regular expressions (given using the syntax of python's `re`-library¹):
 - `[aeiou]+`
 - `\b[^\a]+\b`
 - `\b[^\a]*a[^\a]*a[^\a]*a[^\a]*\b`
 - `[0-9]{6}-[0-9]{4}`
 - `[a-z]{3}[0-9]{3}(?=@alumni.ku.dk)`
- Give a regular expression (using the syntax of python's `re` library) that matches every line of correct RC schemas (as accepted by RC-EVALUATOR; an example is shown in Exercise 3 below).
- Give a lark² grammar for parsing (binary) palindromes, i.e., strings consisting of the characters '0' and '1' that are the same when read left-to-right and right-to-left. For example, "10101", "111111", and "101101101" are palindromes, whereas "1011111" and "101010" are not.

¹<https://docs.python.org/3/library/re.html>

²<https://lark-parser.readthedocs.io/en/stable/>

Exercise 3. *Relational Calculus*

Consider the database schema of a computer shop that consists of four relations, whose schemas are:

```
Product(maker:string, model:int, type:string)
PC(model:int, speed:float, ram:int, hd:int, price:int)
Laptop(model:int, speed:float, ram:int, hd:int, screen:float, price:int)
Printer(model:int, color:bool, type:string, price:int)3
```

Sample data is available on <https://relational-calculus.github.io/RC-Eval/> under the “Computer Shop” example. Manufacturers and model numbers have been “sanitized,” but the data is typical of products on sale at the beginning of 2007.

Write formulas of relational calculus to answer the following queries. Use RC-EVAL to compute the result of your query. Confirm that these results match your intuition when inspecting the data manually. In general, your queries should work for arbitrary data, not just the sample data instance.

- a) What PC models have a ram of 512 MB?
- b) Which manufacturers make laptops with a screen size of 15”?
- c) Find the model number and price of all products (of any type) made by manufacturer E.
- d) Find the makers, model numbers, and prices of all offered devices.
- e) Find the model numbers of all color laser printers that are *not* made by manufacturer E.
- f) Find those manufacturers that sell PCs, but not Printers. Do *not* rely on the correctness of the “type” column in `Product` for this query.
- g) Find those screen sizes that occur in two or more laptops.
- h) Find the manufacturers of Laptops with at least three different hard disk sizes.
- i) Find the manufacturers who sell exactly three different Printer models.
- j) Find those makers that manufacture every type of printer. Do *not* rely on the fact that there are only two types of printers in the given instance.
- k) Validate that the “type” column in `Product` is correct, i.e., only includes correct labels for PCs, Laptops, and Printers with the respective model number. Your query should be a closed formula that outputs the empty set if and only if the correctness is violated.

Exercise 4. *Peer Feedback on Project Idea*

Complete the peer feedback for the two project idea presentations you have been assigned. The feedbackfruit application provides additional instructions on how to structure your feedback along four questions addressing (1) the overall project idea, (2) the used data, (3) the E/R diagram from a syntactic point of view, and (4) the E/R diagram from a semantic point of view.

³RC-EVALUATOR does not support Booleans. We model them as integers, where 1 denotes TRUE and 0 denotes FALSE.